

● MEDIUM

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

05

Q1

ADVANCED MATH

Bacteria are growing in a liquid growth medium. There were 300,000 cells per milliliter during an initial observation. The number of cells per milliliter doubles every 3 hours. How many cells per milliliter will there be 15 hours after the initial observation?

- A. 1,500,000
- B. 2,400,000
- C. 4,500,000
- D. 9,600,000

What is an x -coordinate of an x -intercept of the graph of $y = 3x^2 - 14x + 5$ in the xy -plane?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function $f(t) = 60,000 \cdot 2^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

STUDENT-PRODUCED RESPONSE

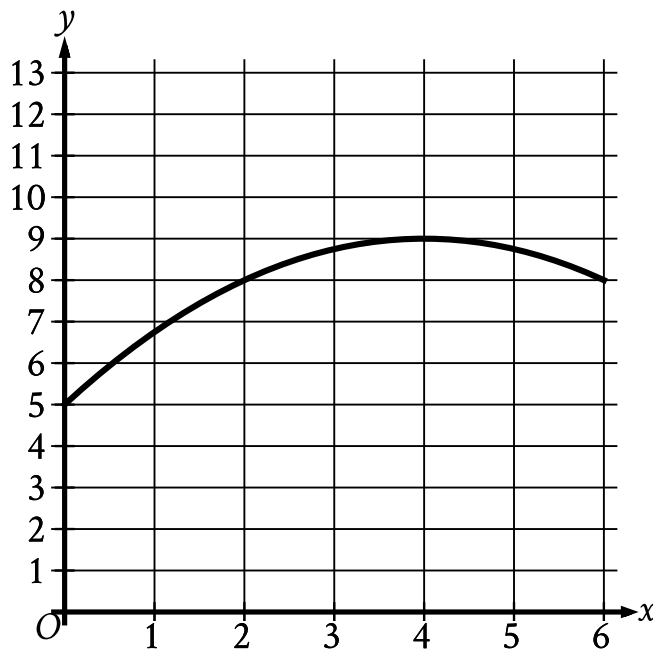
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$D = 5,640(1.9)^t$$

The equation above estimates the global data traffic D , in terabytes, for the year that is t years after 2010. What is the best interpretation of the number 5,640 in this context?

- A. The estimated amount of increase of data traffic, in terabytes, each year
- B. The estimated percent increase in the data traffic, in terabytes, each year
- C. The estimated data traffic, in terabytes, for the year that is t years after 2010
- D. The estimated data traffic, in terabytes, in 2010



- The parabola opens downward.
- The vertex is at point (4 comma 9).
- The parabola passes through the following points:
 - (0 comma 5)
 - (2 comma 8)
 - (4 comma 9)
 - (6 comma 8)

The graph models the number of active projects a company was working on x months after the end of November 2012, where $0 \leq x \leq 6$. According to the model, what is the predicted number of active projects the company was working on at the end of November 2012?

- A. 0
- B. 5

C. 8

D. 9

Q6

ADVANCED MATH

The area A , in square centimeters, of a rectangular painting can be represented by the expression $ww + 29$, where w is the width, in centimeters, of the painting. Which expression represents the length, in centimeters, of the painting?

A. w

B. 29

C. $w + 29$

D. $ww + 29$

$$fx = \frac{a - 19}{x} + 5$$

In the given function f , a is a constant. The graph of function f in the xy -plane, where $y = fx$, is translated 3 units down and 4 units to the right to produce the graph of $y = gx$. Which equation defines function g ?

A. $gx = \frac{a - 19}{x + 4} + 2$

B. $gx = \frac{a - 19}{x - 4} + 2$

C. $gx = \frac{a - 22}{x + 4} + 5$

D. $gx = \frac{a - 22}{x - 4} + 5$

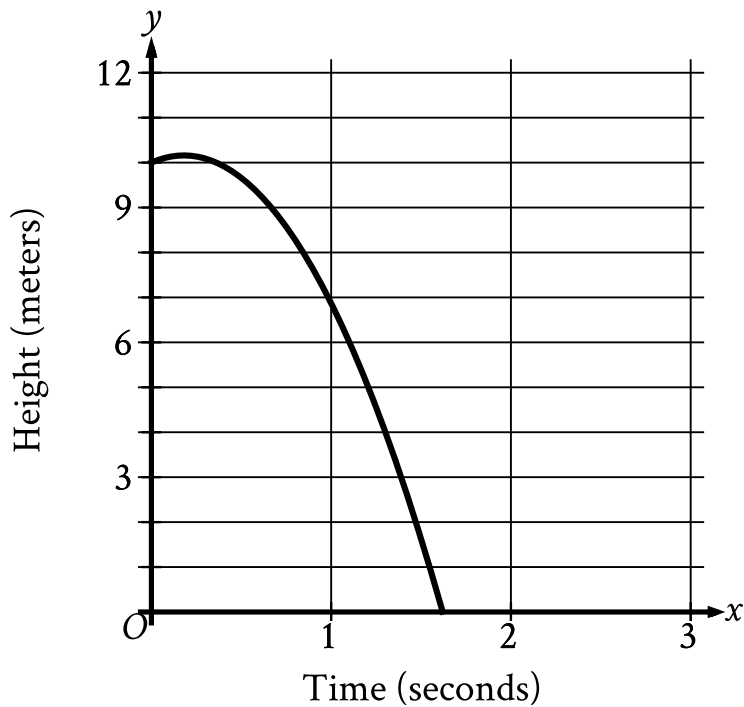
x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 42^x + 3$. In the table, a is a constant. What is the value of a ?

- A. 67
- B. 35
- C. 32
- D. 27

A model predicts that the population of Bergen was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

- A. $p = 0.9615,000^x$
- B. $p = 1.0415,000^x$
- C. $p = 15,0000.96^x$
- D. $p = 15,0001.04^x$



- The parabola opens downward.
- The vertex is at the approximate point (0.2 comma 10.2).
- The parabola passes through the following points:
 - (0 comma 10)
 - approximately (0.2 comma 10.2)
 - approximately (1 comma 6.9)
 - approximately (1.6 comma 0)

A competitive diver dives from a platform into the water. The graph shown gives the height above the water y , in meters, of the diver x seconds after diving from the platform. What is the best interpretation of the x -intercept of the graph?

- A. The diver reaches a maximum height above the water at 1.6 seconds.
- B. The diver hits the water at 1.6 seconds.

- C. The diver reaches a maximum height above the water at 0.2 seconds.
- D. The diver hits the water at 0.2 seconds.